

## Education

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### Doctor of Philosophy – Deep Learning

Mar. 2017 – Aug. 2020

CURTIN UNIVERSITY

Perth, Australia

- Thesis: *Affinity Learning on Graphs with Diffusion Process*
- Awarded the competitive RTP scholarship for international students
- Published five peer-reviewed journal papers and one conference paper
- Graduated with the Chancellor’s Commendation

### Master of Philosophy – Machine Learning

Mar. 2014 – Mar. 2016

CURTIN UNIVERSITY

Perth, Australia

- Thesis: *Clustering by Pairwise Similarity*
- Published two peer-reviewed journal papers and two conference papers
- Graduated with commendation from the Department of Computing

### Bachelor of Science – Computer Science

Sep. 2009 – Jun. 2013

SUN YAT-SEN UNIVERSITY

Guangzhou, China

- Ranked among the top 10 universities in China
- Achieved a course-weighted average grade of 82.6/100

## Employment

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### Senior Lecturer

Jan. 2026 – Present

CURTIN UNIVERSITY

Perth, Australia

- Teaching and research academic, with responsibilities spanning 40% teaching, 40% research, and 20% engagement
- AI Specialisation Coordinator (Bachelor) and AI Major Coordinator in the Master of Computing, leading curriculum design and delivery across the Faculty and international campuses
- Chief Investigator on competitive research grants totaling over \$2 million, including an ARC DECRA, ARC Linkage, CSIRO, and industry partners
- Unit coordinator and lecturer for under- and post-graduate units: *COMP1006 Foundations of Computer Science*, *COMP3010 Machine Learning*, and *COMP6013 Explainable Approaches to Machine Learning*

### AI Specialisation Coordinator (Bachelor)

2025 – Present

CURTIN UNIVERSITY

Perth, Australia

- Coordinated two Bachelor AI Specialisations (SPUC-AINTL and SPUC-AIMLN); led curriculum review in consultation with course leads and unit coordinators across Curtin Bentley and global campuses
- Resolved unit overlap and initiated specialisation merger; improved curriculum coherence and student progression pathways

### AI Major Coordinator – Master of Computing

2025 – Present

CURTIN UNIVERSITY

Perth, Australia

- Coordinated the AI Major within the Master of Computing; oversaw unit offerings, student pathways, and alignment with industry and research priorities
- Collaborated with postgraduate course coordinators to align unit sequencing and strengthen curriculum coherence across the programme

## Lecturer

Dec. 2022 – Dec. 2025

CURTIN UNIVERSITY

Perth, Australia

- Chief Investigator on competitive research grants totaling over \$1.4 million from ARC, CSIRO, and industry partners
- Unit coordinator and lecturer for under- and post-graduate units: *COMP1006 Foundations of Computer Science*, *COMP3010 Machine Learning*, and *COMP6013 Explainable Approaches to Machine Learning*

## Associate Lecturer

Jun. 2020 – Dec. 2022

CURTIN UNIVERSITY

Perth, Australia

- Initiated and led multi-disciplinary research collaborations with colleagues in civil engineering and the humanities, resulting in the co-supervision of two PhD students
- Unit coordinator and lecturer for under- and post-graduate units: *COMP1006 Foundations of Computer Science*, *ISEC2000 Fundamental Concepts of Cryptography*, and *COMP6013 Explainable Approaches to Machine Learning*

## Sessional Academic

Aug. 2018 – Jun. 2020

CURTIN UNIVERSITY

Perth, Australia

- Delivered tutorials, practicals, and laboratory sessions; marked student assessments
- Taught across multiple computing units including *Data Mining*, *Fundamentals of Programming*, *Fundamental Concepts of Cryptography*, and *Artificial and Machine Intelligence*

## R&D Officer

Mar. 2016 – Jan. 2018

RESONANCE HEALTH LTD.

Perth, Australia

- Led a research project on ML-based quantification of liver fat fraction from histological biopsy sections
- Developed CNN models that reduced image processing time from 30 minutes (manual expert review) to under 30 seconds per image

# Teaching

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## COMP3010 Machine Learning

2024–Present

UNDERGRADUATE

Unit Coordinator, Lecturer

- A core unit in the AI major and specialisation, covering foundational machine learning and deep learning
- Coordinated a teaching team of 10 staff across 5 international campuses, supporting over 350 students annually
- Led a comprehensive unit redevelopment in 2024, raising student satisfaction to 87.8%—well above school (75.5%), faculty (78.5%), and university (80.7%) benchmarks

## COMP6013 Explainable Approaches to Machine Learning

2022–Present

POSTGRADUATE

Unit Coordinator, Lecturer

- A core unit in the Master of Artificial Intelligence and Predictive Analytics, focusing on interpretability techniques and ethics in AI/ML
- Designed and developed the unit from scratch; enrolments grew from 11 (2022) to 156 (2025)
- Introduced innovative assessments including VIVA-style oral tests and a machine learning competition on the Kaggle platform, both receiving highly positive feedback

## COMP1006 Foundations of Computer Science

2020–2024

UNDERGRADUATE

Unit Coordinator, Lecturer

- A first-year core unit in computer science, covering discrete mathematics topics such as sets, relations, and graphs
- Coordinated a teaching team of five staff across the Bentley and Miri campuses, supporting approximately 150 students

## ISEC2000 Fundamental Concepts of Cryptography

2021–2023

UNDERGRADUATE

Unit Coordinator, Lecturer

- A second-year core unit in cybersecurity, covering classical and modern cryptographic techniques including DES, RSA, hashing, and digital signatures
- Coordinated a teaching team of three staff across Curtin's Bentley and Miri campuses, supporting approximately 70 students

### Supervision

2020–Present

UNDERGRADUATE & POSTGRADUATE

(Co-)Supervisor

- 2025: Co-supervised a PhD in Humanities, completed on time with five peer-reviewed publications
- 2024: Co-supervised a PhD in Computing, completed on time with three peer-reviewed publications
- Supervised five Honours students and over ten Advanced Science, Capstone, and Master's projects in AI/ML

## Research

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### ARC DECRA – A\$510,000

2026 – 2028

AUSTRALIAN RESEARCH COUNCIL

Principal Investigator

- *Next-Generation Agentic AI System for Intelligent Infrastructure Monitoring*
- Develops an agentic AI system integrating large language models, multi-modal sensing, and autonomous decision-making for continuous, scalable structural health monitoring of civil infrastructure.
- Sole Chief Investigator; responsible for AI architecture design, research direction, and industry and stakeholder engagement.
- One of Australia's most competitive early-career research fellowships, recognising research excellence and independence.

### National Road Safety Action Grant – A\$641,436

Mar. 2025 – Jun. 2027

DEPARTMENT OF INFRASTRUCTURE, TRANSPORT, REGIONAL DEVELOPMENT,  
COMMUNICATIONS AND THE ARTS

Chief Investigator

- *AI-Assisted Design of Sustainable Road Barriers for Improved Road Safety*
- Develops a novel road barrier system using recycled metaconcrete and origami-inspired energy-absorbing structures; ML surrogate models simulate vehicle-barrier collisions under diverse conditions to enable efficient design optimisation.
- Leads the AI/ML component, including development of surrogate models for high-fidelity collision simulations and data-driven design optimisation.
- Developed AI-driven surrogate models that significantly reduced simulation time and enabled fast design iteration aligned with national safety targets.

### ARC Linkage Grant – A\$415,380

Aug. 2024 – Aug. 2027

AUSTRALIAN RESEARCH COUNCIL

Chief Investigator

- *Damage Detection and Quantification using Infrastructure Digital Twins*
- Develops an intelligent structural health monitoring system using digital twins generated from UAV imagery, vibration testing, FEM modelling, and deep learning; enables automated detection and quantification of damage, stiffness loss, and load-carrying capacity.
- Leads the AI and computer vision components, including image-based damage detection, digital twin reconstruction, and structural condition prediction.
- Delivered an integrated AI pipeline for accurate damage detection and quantification, enabling proactive infrastructure maintenance.

## Next Generation AI Graduate Program – A\$350,000

Mar. 2025 – Jun. 2029

CSIRO, WA POLICE

Principal Investigator

- *Next Generation Policing: Effective and Ethical Use of AI and Data Science in Law Enforcement*
- Addresses the growing need for AI and data science capability in policing; supports WA Police in the ethical use of emerging technologies for data linkage, visualisation, and insight generation; trains seven Master's and five Honours students as multidisciplinary data specialists.
- Leads the project and coordinates stakeholders across CSIRO, WA Police, Curtin, UWA, ECU, and UNE; supervises students and manages cross-institutional collaboration.
- Established a cross-sector collaboration, launched an AI ethics and capability framework, and initiated the first training cohort.

## Resources Technology and Critical Minerals Trailblazer EMCR Grant – A\$20,000

Jun. 2025 – Jun. 2026

CURTIN UNIVERSITY

Principal Investigator

- *AI-Empowered Multi-Modality Digital Twin for Reclaimers*
- Develops an AI-driven digital twin system for structural monitoring of reclaimer machines, integrating drone imaging, IoT sensor data, and defect detection models for real-time condition assessment and predictive maintenance.
- Leads the project, designs the AI pipeline, oversees drone and sensor integration, and coordinates field validation with industry partners.
- Supports innovation in critical minerals infrastructure through intelligent automation, aligned with UN SDG 9 (Industry, Innovation and Infrastructure).

## Publications

### Top 10 Publications

- [1] **Li, Qilin**, Y. Wang, W. Chen, L. Li, and H. Hao. "Machine learning prediction of BLEVE loading with graph neural networks". In: *Reliability Engineering & System Safety* 241 (2024), p. 109639. **(Q1, IF 9.4, top journal in engineering safety)**.
- [2] **Li, Qilin**, Z. Wang, L. Li, H. Hao, W. Chen, and Y. Shao. "Machine learning prediction of structural dynamic responses using graph neural networks". In: *Computers & Structures* 289 (2023), p. 107188. **(Q1, IF 4.4)**.
- [3] **Li, Qilin**, Z. Wang, W. Chen, L. Li, and H. Hao. "Advancing blast fragmentation simulation of RC slabs: A graph neural network approach". In: *Engineering Structures* 308 (2024), p. 118009. **(Q1, IF 5.6, top journal in structural engineering)**.
- [4] **Li, Qilin**, Y. Wang, Y. Shao, L. Li, and H. Hao. "A comparative study on the most effective machine learning model for blast loading prediction: From GBDT to Transformer". In: *Engineering Structures* 276 (2023), p. 115310. **(Q1, IF 5.6, top journal in structural engineering)**.
- [5] **Li, Qilin**, S. An, L. Li, W. Liu, and Y. Shao. "Multi-view diffusion process for spectral clustering and image retrieval". In: *IEEE Transactions on Image Processing* 32 (2023), pp. 4610–4620. **(Q1, IF 10.8, top journal in computer vision)**.
- [6] **Li, Qilin**, S. An, W. Liu, and L. Li. "Semisupervised learning on graphs with an alternating diffusion process". In: *IEEE Transactions on Neural Networks and Learning Systems* 32.7 (2020), pp. 2862–2874. **(Q1, IF 10.2, top journal in machine learning)**.
- [7] **Li, Qilin**, W. Liu, and L. Li. "Affinity learning via a diffusion process for subspace clustering". In: *Pattern Recognition* 84 (2018), pp. 39–50. **(Q1, IF 7.5, top journal in pattern recognition)**.
- [8] J. Liu, **Li, Qilin**, L. Li, and S. An. "Structural damage detection and localization via an unsupervised anomaly detection method". In: *Reliability Engineering & System Safety* 252 (2024), p. 110465. **(Q1, IF 9.4, top journal in engineering safety)**.
- [9] Y. Shao, L. Li, J. Li, **Li, Qilin**, S. An, and H. Hao. "Out-of-plane full-field vibration displacement measurement with monocular computer vision". In: *Automation in Construction* 165 (2024), p. 105507. **(Q1, IF 9.6, top journal in construction automation)**.

- [10] J. Li, **Li, Qilin**, H. Hao, and L. Li. “Prediction of BLEVE blast loading using CFD and artificial neural network”. In: *Process Safety and Environmental Protection* 149 (2021), pp. 711–723. (**Q1, IF 6.9**).

### Other Publications in 2025-2023

- [11] **Li, Qilin**, L. Li, Y. Shao, R. Wang, and H. Hao. “A multi-task machine learning approach for data efficient prediction of blast loading”. In: *Engineering Structures* 326 (2025), p. 119577. (**Q1, IF 5.6, top journal in structural engineering**).
- [12] **Li, Qilin**, Y. Shao, L. Li, J. Li, and H. Hao. “Advancements in 3D displacement measurement for civil Structures: A monocular vision approach with moving cameras”. In: *Measurement* 242 (2025), p. 116060. (**Q1, IF 5.2**).
- [13] Y. Shao, L. Li, J. Li, **Li, Qilin**, S. An, and H. Hao. “Robust 3D vessel trajectory monitoring with monocular vision based deep learning and surveillance cameras”. In: *Ocean Engineering* 332 (2025), p. 121429. (**Q1, IF 4.6**).
- [14] Y. Shao, L. Li, J. Li, **Li, Qilin**, S. An, and H. Hao. “DIMMC: A 3D vision approach for structural displacement measurement using a moving camera”. In: *Engineering Structures* 338 (2025), p. 120566. (**Q1, IF 5.6, top journal in structural engineering**).
- [15] Z. Wang, **Li, Qilin**, W. Chen, H. Hao, and L. Li. “A probability-based risk assessment of secondary fragments ejected from the reinforced concrete wall under close-in explosions”. In: *Structural Safety* 114 (2025), p. 102565. (**Q1, IF 5.7**).
- [16] Y. Zhu, **Li, Qilin**, W. Liu, and C. Yin. “Diffusion process with structural changes for subspace clustering”. In: *Pattern Recognition* 158 (2025), p. 111066. (**Q1, IF 7.5, top journal in machine learning**).
- [17] Y. Zhu, **Li, Qilin**, W. Liu, C. Yin, and Z. Gao. “A Novel Exploration of Diffusion Process Based on Multi-Type Galton–Watson Forests”. In: *Mathematics* 12.22 (2024), p. 3462. (**Q1, IF 2.3**).
- [18] R. Wang, Y. Shao, **Li, Qilin**, L. Li, J. Li, and H. Hao. “A novel transformer-based semantic segmentation framework for structural condition assessment”. In: *Structural Health Monitoring* 23.2 (2024), pp. 1170–1183. (**Q1, IF 5.7**).
- [19] Q. Hu, X. Zhang, **Li, Qilin**, H. Hao, C. Coffey, and F. Mitchell-Corbett. “Prediction and interpretability of accidental explosion loads from hydrogen-air mixtures using CFD and artificial neural network method”. In: *International Journal of Hydrogen Energy* 66 (2024), pp. 135–147. (**Q1, IF 8.1**).
- [20] R. Cheng, W. Chen, H. Hao, and **Li, Qilin**. “Prediction of BLEVE-induced response of road tunnel using Transformer network with modified self-attention (SAMT)”. In: *Engineering Structures* 314 (2024), p. 118415. (**Q1, IF 5.6, top journal in structural engineering**).
- [21] Y. Shao, L. Li, J. Li, **Li, Qilin**, S. An, and H. Hao. “3DGEN: a framework for generating custom-made synthetic 3D datasets for civil structure health monitoring”. In: *Structural Health Monitoring* (2024), p. 14759217241265540. (**Q1, IF 5.7**).
- [22] R. Wang, J. Li, L. Li, S. An, B. Ezard, **Li, Qilin**, and H. Hao. “Structural damage identification by using physics-guided residual neural networks”. In: *Engineering Structures* 318 (2024), p. 118703. (**Q1, IF 5.6, top journal in structural engineering**).
- [23] Y. Shao, L. Li, J. Li, **Li, Qilin**, S. An, and H. Hao. “3D displacement measurement using a single-camera and mesh deformation neural network”. In: *Engineering Structures* 318 (2024), p. 118767. (**Q1, IF 5.6, top journal in structural engineering**).
- [24] Z. Wang, **Li, Qilin**, W. Chen, H. Hao, and L. Li. “Fragment prediction of reinforced concrete wall under close-in explosion using Fragment Graph Network (FGN)”. In: *Computers & Structures* 305 (2024), p. 107556. (**Q1, IF 4.4**).
- [25] J. Liu, **Li, Qilin**, S. An, B. Ezard, and L. Li. “EdgeConvFormer: An Unsupervised Anomaly Detection Method for Multivariate Time Series”. In: *International Conference on Pattern Recognition*. Springer Nature Switzerland Cham. 2024, pp. 367–382.

- [26] T. Xie, H. Chen, W. Liu, R. Zhou, and **Li, Qilin**. “3D surface segmentation from point clouds via quadric fits based on DBSCAN clustering”. In: *Pattern Recognition* 154 (2024), p. 110589. (**Q1, IF 7.5, top journal in pattern recognition**).
- [27] **Li, Qilin**, Y. Wang, L. Li, H. Hao, R. Wang, and J. Li. “Prediction of BLEVE loads on structures using machine learning and CFD”. In: *Process Safety and Environmental Protection* 171 (2023), pp. 914–925. (**Q1, IF 6.9**).
- [28] W. Zheng, J. Li, **Li, Qilin**, and H. Hao. “Multi-channel response reconstruction using transformer based generative adversarial network”. In: *Earthquake Engineering & Structural Dynamics* 52.11 (2023), pp. 3369–3391. (**Q1, IF 4.3**).
- [29] Y. Shao, L. Li, J. Li, **Li, Qilin**, S. An, and H. Hao. “Monocular vision based 3D vibration displacement measurement for civil engineering structures”. In: *Engineering Structures* 293 (2023), p. 116661. (**Q1, IF 5.6, top journal in structural engineering**).
- [30] T. Malatesta, **Li, Qilin**, J. K. Breadsell, and C. Eon. “Distinguishing household groupings within a precinct based on energy usage patterns using machine learning analysis”. In: *Energies* 16.10 (2023), p. 4119.

### Other Publications before 2023

| Journal / Conference                                                                      | No. of Publications |
|-------------------------------------------------------------------------------------------|---------------------|
| Q1 machine learning journals (e.g., <i>Pattern Recognition</i> , <i>Neuro-computing</i> ) | 4                   |
| Other applied AI journals (e.g., <i>Multimedia Tools and Applications</i> )               | 3                   |
| <i>IEEE Conference Proceedings</i> (e.g., DICTA, FSKD)                                    | 4                   |
| <b>Total</b>                                                                              | <b>11</b>           |

## Engagement

### Major Internal Engagement Roles

#### Multi-Disciplinary Researcher

2020 – Present

CURTIN UNIVERSITY

University-level

- Applied AI to interdisciplinary challenges through collaborations with the Centre for Infrastructural Monitoring and Protection (CIMP) and the Curtin University Sustainability Policy (CUSP) Institute
- Led to two major grants (ARC Linkage and National Road Safety Action), co-supervision of five PhD students (two completed, three ongoing), and over 20 peer-reviewed publications

#### Next Generation AI Graduate Program Coordinator

2025 – Present

FACULTY OF SCIENCE AND ENGINEERING, CURTIN UNIVERSITY

Faculty-level

- Coordinated the program across internal stakeholders including the ARC Training Centre for Transforming Maintenance through Data Science, the Curtin Institute for Data Science, and the Future of Work Institute; aligned student recruitment with project and supervisor priorities
- Strengthened AI visibility across the Faculty and built cross-institutional networks in AI and data science

#### Curtin Global Campuses Connection Lead

2020 – Present

CURTIN UNIVERSITY

University-level

- Collaborated with Curtin’s global campus colleagues on iSoLT teaching innovation projects; contributed to curriculum design for *Machine Learning*, *Foundations of Computer Science*, and *Fundamental Concepts of Cryptography*
- Secured the Curtin Academy iSoLT Teaching Innovation Grant (2024); improved cross-campus unit delivery and alignment

### Major External Engagement Roles

## Demand-Driven Researcher

2023 – Present

AUSTRALIA

National-level

- Conducted demand-driven research with government agencies, industry, and academia to address real-world challenges in AI and data science
- Secured major grants (National Road Safety Action Grant, ARC Linkage, CSIRO Graduate Program); established HDR scholarships, internships, and industry engagement pathways

## International Universities Partnership Facilitator

2024 – Present

GLOBAL

International-level

- Taught COMP1005 *Fundamentals of Programming* in the Curtin–Wuhan Institute of Technology program during a 10-day visit (June 2024); served as external reviewer for the Teaching Development Grant at City University of Hong Kong
- Strengthened Curtin’s international academic partnerships and promoted global collaboration in curriculum innovation

## Discipline Leadership and Editorial Engagement

2020 – Present

GLOBAL

International-level

- Served on the organising committee of the International Conference on Infrastructural Monitoring and Protection; Guest Editor for *Mathematics* and *Buildings*; Editorial Board Member, *Urban Lifeline*; regular reviewer for over 20 Q1 journals

## Honors & Awards

|      |                                                                                                                                                                    |                             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 2026 | <b>ARC DECRA – A\$510,000</b> , ARC Discovery Early Career Researcher Award for <i>Next-Generation Agentic AI System for Intelligent Infrastructure Monitoring</i> | Australian Research Council |
| 2025 | <b>Trailblazer EMCR Grant</b> , Curtin Trailblazer Early-Career Researcher Grant in recognition of research excellence                                             | Curtin University           |
| 2024 | <b>Teaching Innovation Grant</b> , Curtin Academy iSoLT Teaching Innovation Grant for <i>Interactive Gamified Learning Platform for Curtin Global Campuses</i>     | Curtin University           |
| 2023 | <b>Third Prize – ICSHM 2022 Competition</b> , Third prize in the international Structural Health Monitoring competition (ICSHM 2022) among 70+ global teams        | International               |
| 2020 | <b>PhD Graduation Commendation</b> , Chancellor’s Commendation for outstanding academic performance upon PhD graduation                                            | Curtin University           |
| 2017 | <b>RTP Scholarship</b> , Competitive Research Training Program (RTP) Scholarship for PhD study as an international student                                         | Curtin University           |
| 2016 | <b>MPhil Graduation Commendation</b> , Commendation from the Department of Computing for academic excellence in the MPhil program                                  | Curtin University           |

## Academic Service

### Organising Committee Member

Jan. 2025 – Dec. 2025

CIMP1-2025

Perth, Australia

- Serving on the scientific organising committee for the inaugural International Conference on Infrastructural Monitoring and Protection

### Invited Guest Editor

Apr. 2025 – Oct. 2025

*Buildings*

- Special Issue: *AI-Powered Structural Health Monitoring: Innovations and Applications*

### Invited Guest Editor

Aug. 2024 – Mar. 2026

*Mathematics*

- Special Issue: *Advances in Deep Learning, Computer Vision, and Engineering Applications*

## Editorial Board Member

*Jul. 2024 – Jul. 2026*

*Urban Lifeline*

- Early Career Editorial Board Member

## Reviewer

*2016 – Present*

Q1 JOURNALS IN AI, ML, SHM, AND STRUCTURAL ENGINEERING

- Regular reviewer for 20+ Q1 journals across AI, machine learning, structural health monitoring, and structural engineering